Predictive Lead Scoring Using Machine Learning

Optimizing Sales ROI by transforming dormant wallet holders into active transactors

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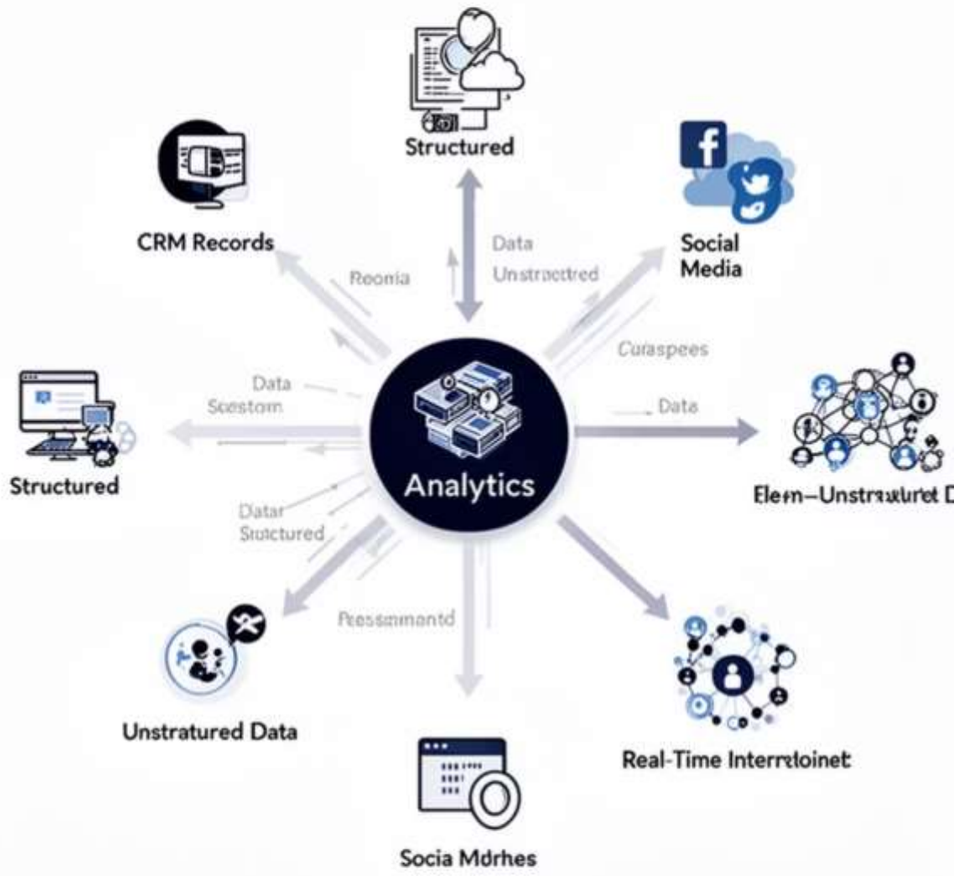


Introduction — Why change matters

The Evolutionary Shift

Transitioning from manual, rule-based point systems to sophisticated predictive models driven by empirical historical data.

Traditional Flaws

Static criteria assumed linear customer paths, leading to "blind" marketing and high churn rates.

Modern ML Approach

Algorithms analyze thousands of variables simultaneously, including onboarding navigation speed and specific registration channels.

The Predictive Goal

Moving from descriptive analytics (what happened) to predictive action (what will happen) via a calculated "Propensity Score".

Business Problem & Solution Overview

The Dormancy Gap

Identifying high-value customers who completed registration and set a PIN but failed to reach the "Activation" stage (first transaction)

Operations Pain

Indiscriminate "broadcast" campaigns waste marketing budgets on leads with low statistical probability of reactivation.

Proposed Strategy

Score users with a Reactivation Propensity and trigger targeted incentives (high-value SMS, tailored discounts) for high-score leads.

Analytical Techniques & Data Engineering

We built a robust transformation layer in SQL (Metabase) to centralize feature engineering and accelerate model iteration.

Custom SQL Engineering

Developed a transformation layer in Metabase to perform "Feature Engineering" directly at the database level.

Data Hygiene

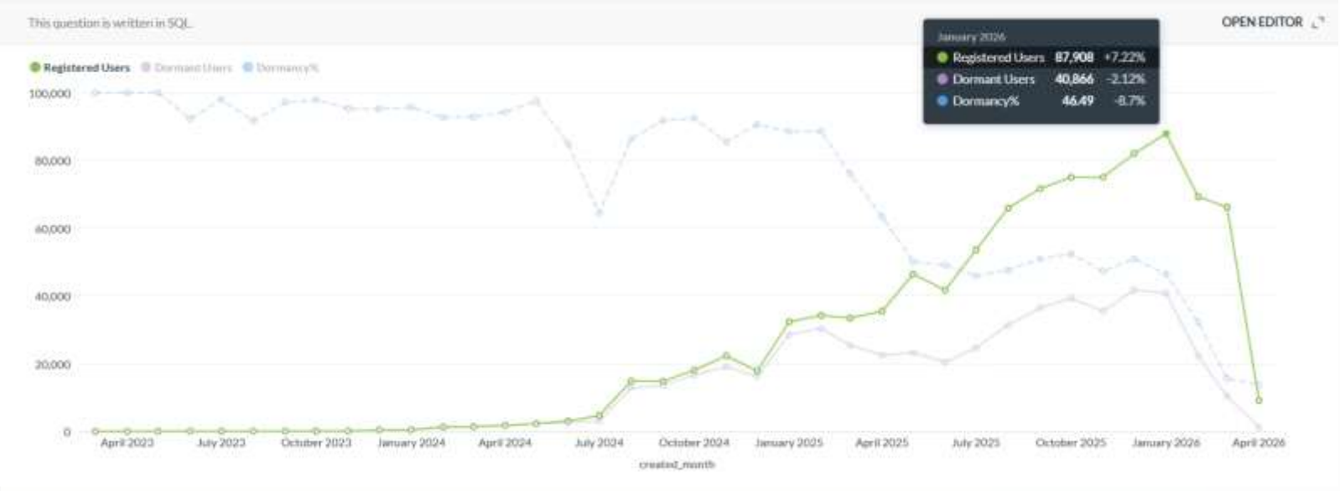
Automated filters remove suspended/high-risk accounts and transactional noisy to protect model quality.

Raw Data Isolation

Focused on fields like `Wallet Createdat Local` and `Wallet Lasttxnts Local` to define the dormancy window

Dormancy SQL Code & Visualization

```sql
WITH base AS (
    SELECT
        w.walletid,
        DATE_TRUNC('month', w.wallet_createdat_local) AS created_month,
        w.wallet_createdat_local,
        w.wallet_lasttxnts_local AS last_txn_date
    FROM "dbt-dimensions"."wallets_dimension" w
    WHERE w.wallet_status = 'WalletStatus_ACTIVE'
      AND w.pinsetflag = TRUE
),

classified AS (
    SELECT
        walletid,
        created_month,
        CASE
            WHEN last_txn_date IS NULL
              OR last_txn_date > wallet_createdat_local + INTERVAL '30 days'
            THEN 1 ELSE 0
        END AS is_dormant
    FROM base
)

SELECT
    created_month,
    COUNT(DISTINCT walletid) AS created_users,

    COUNT(DISTINCT CASE WHEN is_dormant = 1 THEN walletid END) AS dormant_users,

    -- 📊 Percentage
    ROUND(
        COUNT(DISTINCT CASE WHEN is_dormant = 1 THEN walletid END) * 100.0
        / COUNT(DISTINCT walletid),
    2) AS dormancy_rate_pct

FROM classified
GROUP BY 1
ORDER BY 1;
```

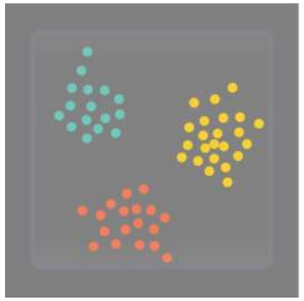


Cohort SQL Code & Analysis

reg_month	channel	total_registered	m0	m1	m2	m3
January 1, 2026, 12:00 AM	Fawry	65,107	61,376	23,212	19,816	9,584
January 1, 2026, 12:00 AM	Retail	25,686	21,764	16,108	15,288	3,337
February 1, 2026, 12:00 AM	Fawry	46,055	41,942	14,356	6,738	0
February 1, 2026, 12:00 AM	Retail	25,992	22,866	19,942	3,222	0
March 1, 2026, 12:00 AM	Fawry	41,846	38,106	8,613	0	0
March 1, 2026, 12:00 AM	Retail	26,686	22,667	4,554	0	0
April 1, 2026, 12:00 AM	Fawry	7,053	6,241	0	0	0
April 1, 2026, 12:00 AM	Retail	3,137	1,765	0	0	0

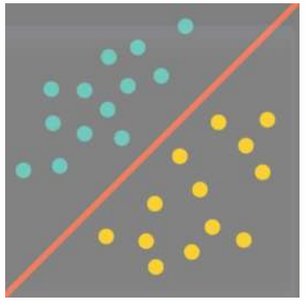
```sql
WITH wallets AS (
    SELECT
        w.walletid,
        CASE
            WHEN w.registeredby != '8c045c06-1f13-4667-bd42-c4936e1b029c'
            THEN 'Retail'
            ELSE 'Fawry'
        END AS channel,
        DATE_TRUNC('month', w.wallet_registeredat_local) AS reg_month
    FROM "dbt-dimensions"."wallets_dimension" w
    WHERE w.currentflag = TRUE
      AND w.wallet_type = 'WalletType_CONSUMER'
      AND w.wallet_registeredat_local >= DATE '2026-01-01'
      AND w.wallet_registeredat_local < DATE '2026-05-01'
)

SELECT
    w.reg_month,
    w.channel,

    COUNT(DISTINCT w.walletid) AS total_registered,

    COUNT(DISTINCT CASE
        WHEN DATE_TRUNC('month', t.transaction_createdat_local) = w.reg_month
        THEN w.walletid END) AS m0,

    COUNT(DISTINCT CASE
        WHEN DATE_TRUNC('month', t.transaction_createdat_local) = w.reg_month + INTERVAL '1 month'
        THEN w.walletid END) AS m1,

    COUNT(DISTINCT CASE
        WHEN DATE_TRUNC('month', t.transaction_createdat_local) = w.reg_month + INTERVAL '2 month'
        THEN w.walletid END) AS m2,

    COUNT(DISTINCT CASE
        WHEN DATE_TRUNC('month', t.transaction_createdat_local) = w.reg_month + INTERVAL '3 month'
        THEN w.walletid END) AS m3

FROM wallets w
LEFT JOIN "dbt-dimensions"."transactions_dimension" t
    ON w.walletid = t.walletdetailsid
   AND t.currentflag = TRUE
   AND t.transactionstatus IN (
        'TransactionStatus_POSTED',
        'TransactionStatus_POSTED_FAWRY'
    )
    AND t.transaction_createdat_local >= DATE '2026-01-01'
    AND t.transaction_createdat_local < DATE '2026-05-01'

GROUP BY 1,2
ORDER BY 1,2;
```

AI Solution: Algorithmic Architecture



Ensemble Models (Random Forest & XGBoost)

Primary drivers of the Reactivation Score, capable of handling non-linear fintech behavioral patterns.



Unsupervised Learning (K-Means Clustering)

Identifies "Dormancy Personas," such as users stuck at the PIN-setting stage.

Reinforcement Learning

Using models to determine if a user's risk level or registration source is the top predictor of activity.

Digital Transformation Impact

Intelligence Gate

Automated prioritization replaces manual lists—only high-propensity leads receive resource-intensive outreach.

Process Efficiency

Eliminated repetitive tasks (manual matching, spreadsheets), enabling the team to focus on creative campaign design.

Cultural Shift

Decisions guided by statistical evidence rather than intuition; continuous measurement becomes standard practice.

Dynamic Prioritization

Replaced simple First-In, First-Out (FIFO) methods with intent-based prioritization

ERP & Operational Integration

Scores flow from the ML layer into MS Dynamics 365 via API; ERP transactional logs feed back into the SQL feature store—creating a closed loop.

1

1. Transactional Truth

Local ERP as financial source of truth to attribute revenue to campaigns.

2

2. Bidirectional Sync

APIs push scores to CRM; successful activations are written back to ERP for validation and crediting.

3

3. Performance Attribution

Revenue from re-activated users directly tied to ML-driven outreach for ROI measurement.

4

4. Closed-Loop Loop

Successful transactions are recorded back in the ERP, validating the AI's prediction.

Business Impact & Value

40%

Retention Uplift

Targeted ML users remained active 40% more in month two versus generic outreach cohorts.

Seconds

Lead Identification Speed

Actionable leads surfaced in seconds after data ingestion down from days with manual methods.

Reduced

Ad Spend Waste

Precision targeting narrows the audience, significantly reducing wasted impressions and cost-per-acquisition.

Strategic advantage: rescuing at-risk customers pre-churn and demonstrating measurable ROI tied to ERP revenue records.

Risks, Ethics & Limitations

- **Data Privacy** — Hash columns for identifiers; GDPR and local banking compliance considered in data handling and retention.
- **Algorithmic Bias** — Continuous audits of features (e.g., RiskLevel) to detect discriminatory signals and recalibrate models.
- **Model Drift & Fragility** — Monitoring and retraining cadence to maintain predictive performance; fallback rules if APIs fail.
- **Security Standards** — Operational controls aligned with ISO 27001 Clause 8.1 and secure deployment practices.

Recommendations & Future Scope

1

Short Term — Real-Time Nudges

Move from batch to streaming to deliver instant in-app nudges when users show activation intent.

2

Mid Term — NLP Insights

Analyze CRM complaint logs to separate technical issues from price objections and route remediation automatically.

3

Mid/Long Term — Predictive Cross-Sell

Extend re-activation logic to identify micro-loan and insurance candidates, increasing lifetime value.

4

Long Term — Sequence Models

Experiment with RNNs/transformers to capture sequential intent and anticipate churn patterns earlier.